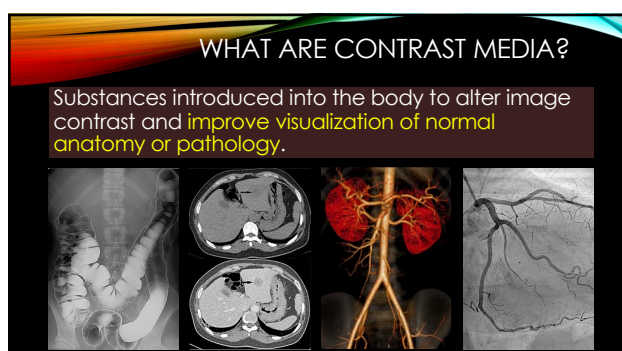




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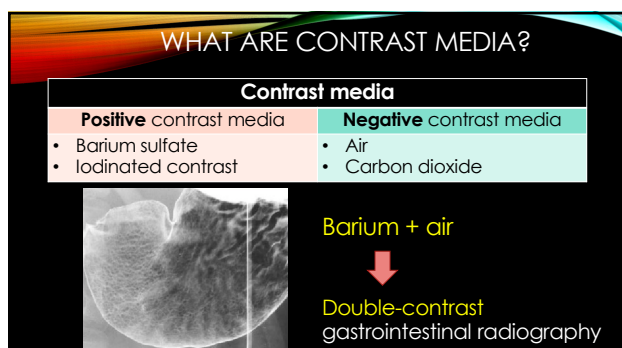
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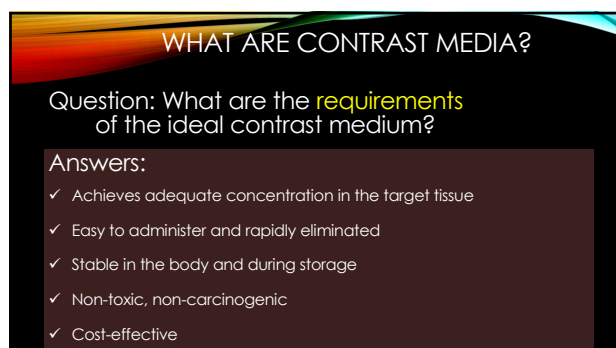
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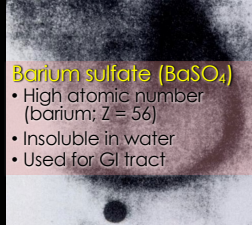
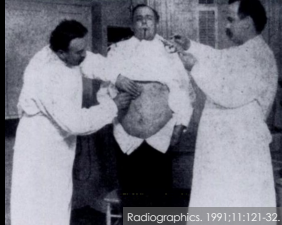
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WHAT ARE CONTRAST MEDIA?

The first contrast study on the upper gastrointestinal tract was performed in the early 1900s.



Barium sulfate (BaSO_4)

- High atomic number (barium; $Z = 56$)
- Insoluble in water
- Used for GI tract

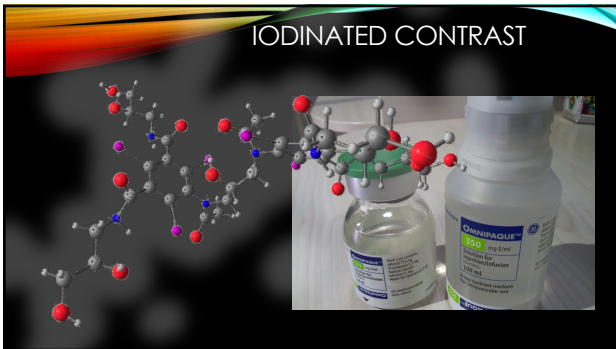
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 Adverse effects of iodinated contrast

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IODINATED CONTRAST



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IODINATED CONTRAST

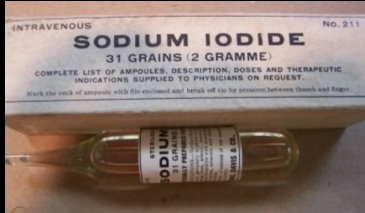
Compounds containing covalently bound **iodine** that increase X-ray attenuation.



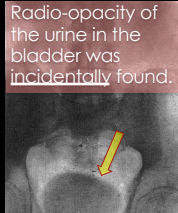
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IODINATED CONTRAST

In the early 1920s, syphilis was treated with **sodium iodide (ionic iodide!)**.



Radio-opacity of the urine in the bladder was incidentally found.



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TOXICITY OF IODINE

Acute symptoms that may develop when an excess amount of **INORGANIC iodine** is administered intravenously.

- Nausea and vomiting
- Diarrhea (blood in stool, in some cases)
- Fever
- Headache
- Pulmonary edema
- Delirium
- Stupor
- Shock



Institute of Medicine, Food and Nutrition Board
Washington, DC: National Academy Press, 2001.

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TOXICITY OF IODINE

• **Dietary reference** intakes for iodine

Age	Male	Female	Pregnancy	Lactation
Birth to 6 months	110 μg	110 μg		
7–12 months	130	130		
1–3 years	90	90		
4–8 years	90	90		
9–13 years	120	120		
14–18 years	150	150	220 μg	290 μg
19+ years	150	150	220	290

Institute of Medicine, Food and Nutrition Board, Washington, DC: National Academy Press, 2001.

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TOXICITY OF IODINE

Question: How much iodine is administered during a routine contrast-enhanced CT examination?

Answer: approx. 30 g.

$$300 \text{ mgI/mL} \times 2 \text{ mL/kg-body weight} \times 50 \text{ kg} = 30,000 \text{ mgI}$$



=



Why can this be administered **safely**?

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Toward safer contrast

Physics of iodinated contrast

Pharmacokinetics of contrast media

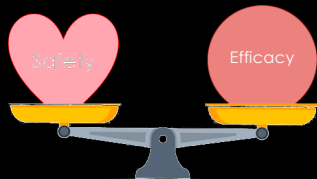
Applications of iodinated contrast

Adverse effects of iodinated contrast

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TOWARD SAFER CONTRAST

We need a contrast agent that is ...
safe to administer and **effective at enhancing**.

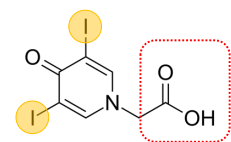


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TOWARD SAFER CONTRAST

The first-generation iodine-based contrast media.
Pyridine ring with two iodine atoms attached.

Carboxyl group (-COOH)
Added to make it
SOLUBLE.

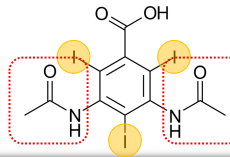


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TOWARD SAFER CONTRAST

Early modern iodinated contrast media were based on a triiodinated benzene ring.
 Tri-iodobenzoic acid carries three iodine atoms
 --> **more radiopaque**

Acetamido side groups were added to improve **water solubility** and **reduce protein binding and toxicity**.



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TOWARD SAFER CONTRAST

- However, the agents still caused adverse effects, due to their "**high osmolality**".

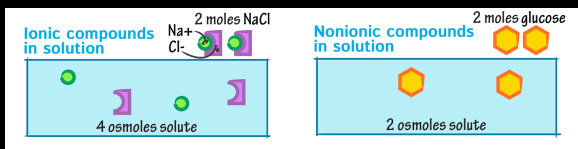
Why does osmolality matter?

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TOWARD SAFER CONTRAST

Question: What is OSMOLALITY?

Answer: The number of solute particles dissolved in a unit weight of water.



<https://www.drawitoknowit.com/course/biochemistry/glossary/cellular-anatomy-physiology/osmolality>

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TOWARD SAFER CONTRAST



The normal osmolality of human blood plasma.

≈ **290 mOsm/kg**



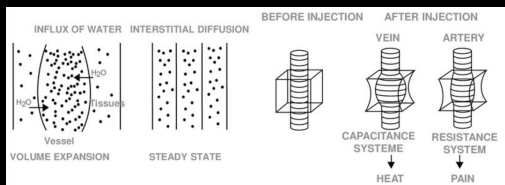
The osmolality of **IONIC** iodinated contrast media.

≈ **1200-2400 mOsm/kg!**

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TOWARD SAFER CONTRAST

Osmotic activity on the body



There remained a need to find a **water-soluble** but **lower osmolality** (less toxic) contrast agent.

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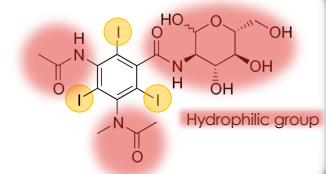
TOWARD SAFER CONTRAST

The first **non-ionic** contrast medium (1974).



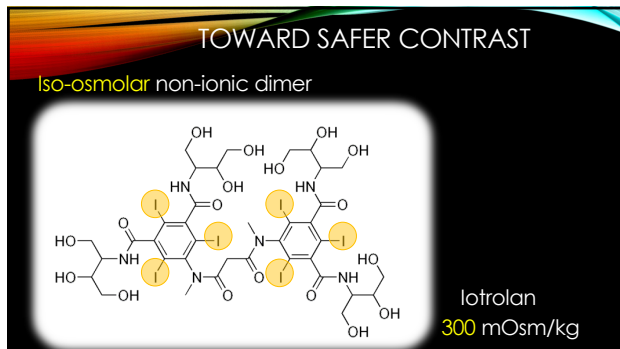
Torsten Almén, MD (1931 - 2016)

"The father of non-ionic iodine contrast media"



Metrizamide
485 mOsm/kg

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TOWARD SAFER CONTRAST

Summary - the chemical structures of the iodinated contrast

Ionization	Ionic	Non-ionic
Monomers		
Example	Diatrizoate (Hypaque®) Iothalamate (Conray®)	Iopamidol (Iopamiro®) Iopromide (Ultravist®) Iohexol (Omnipaque®) Ioversol (Optiray®) Iobitridol (Xenetic®)
Dimers		
Example	Ioxaglate (Hexabrix®)	Iodixanol (Visipaque®)

Ionic contrast media are rarely used intravascularly (but still used for some GI examinations)

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PHYSICS OF IODINATED CONTRAST

- The ability to distinguish between tissues of different x-ray attenuation (image contrast) depends on two types of interactions between photons and matter:
 - Photoelectric absorption:** depends on both physical density and *atomic number*.
 - Compton scattering:** depends primarily on electron density.

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PHYSICS OF IODINATED CONTRAST

Photoelectric effect

- Occurs when an incident X-ray photon has enough energy to eject an inner-shell electron and is completely absorbed.
- Relatively **important at lower photon energies and in higher-Z materials** such as iodine and bone.

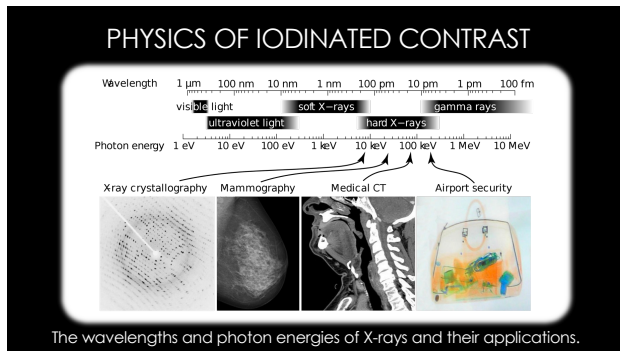
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PHYSICS OF IODINATED CONTRAST

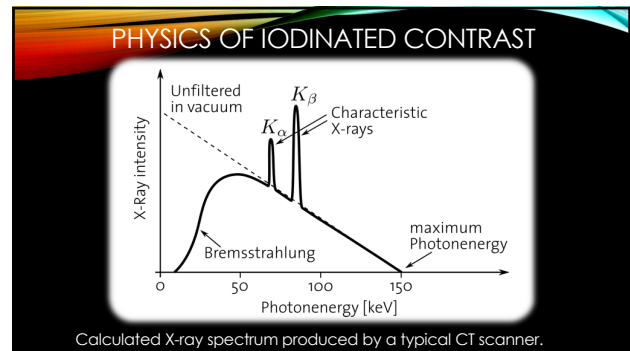
Compton scattering

- Occurs when the photon energy is much greater than the electron binding energy.
- The photon transfers part of its energy and momentum to an electron and is scattered with lower energy.
- Dominant interaction in soft tissue over much of the diagnostic X-ray energy range.

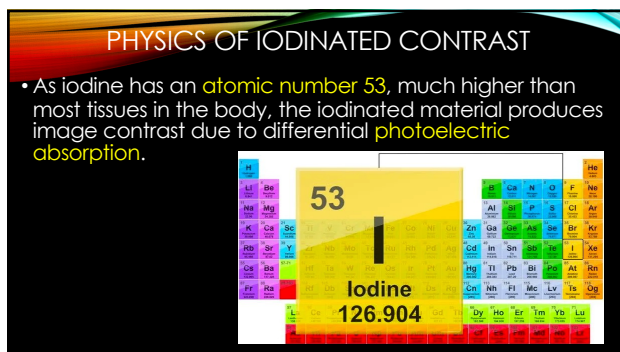
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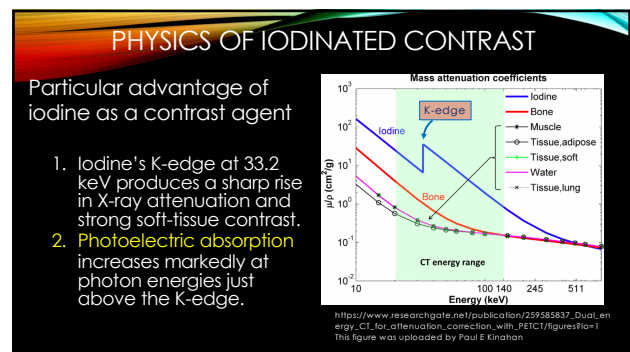
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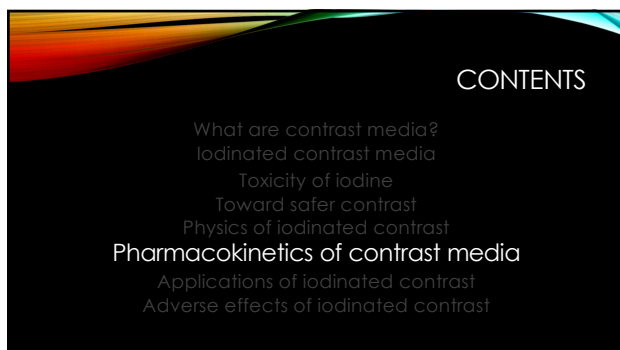
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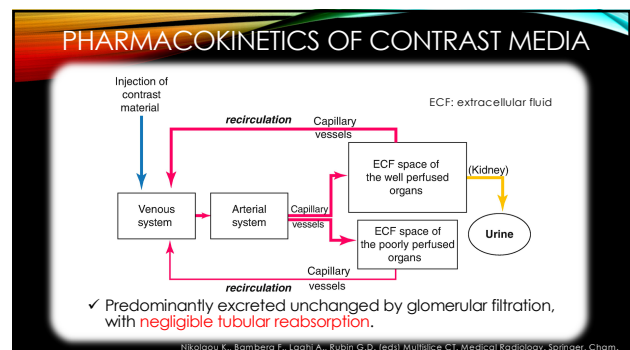
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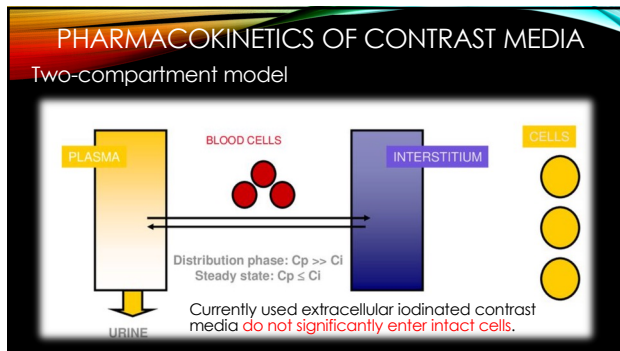
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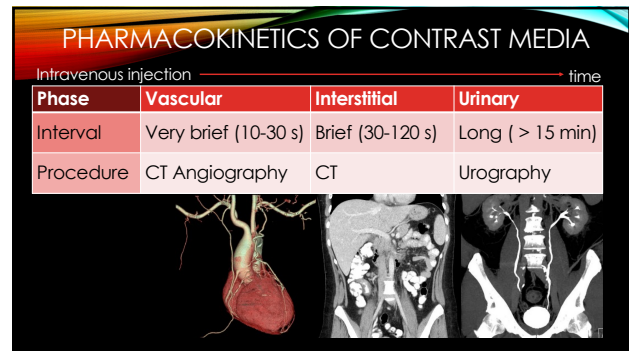
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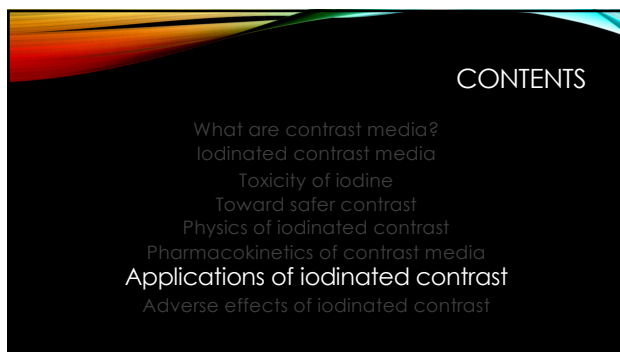
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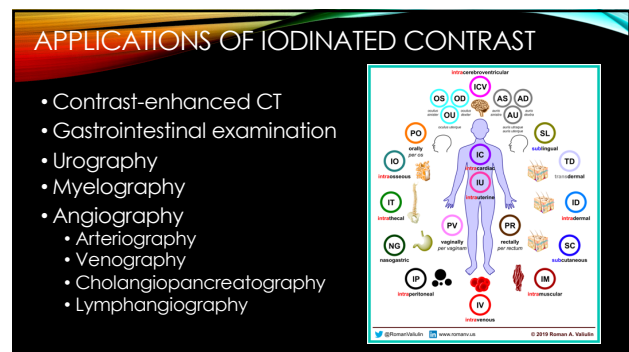
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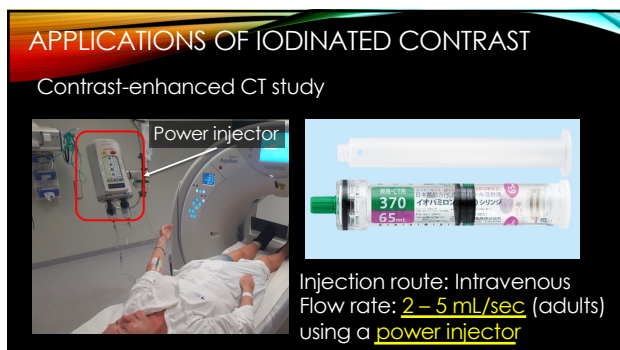
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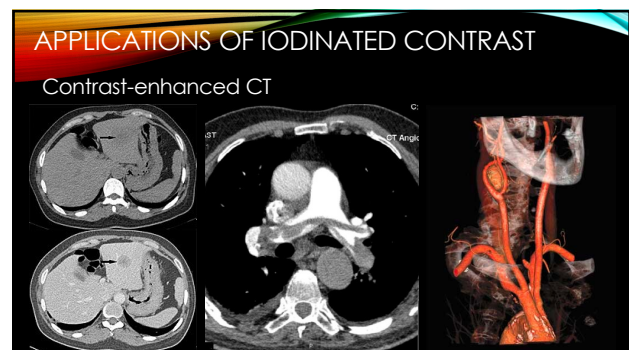
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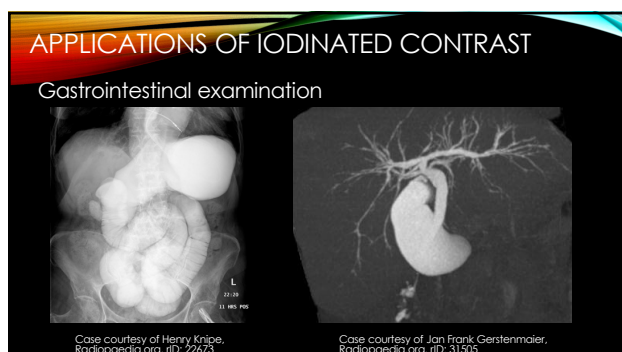
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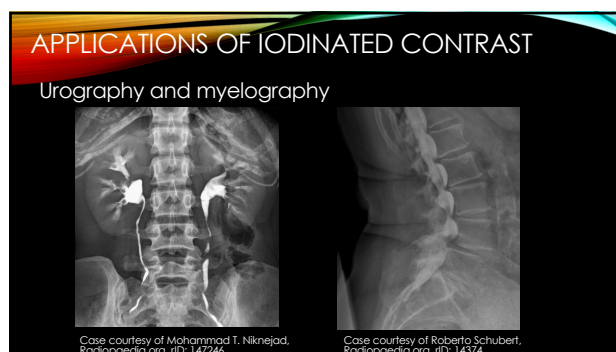
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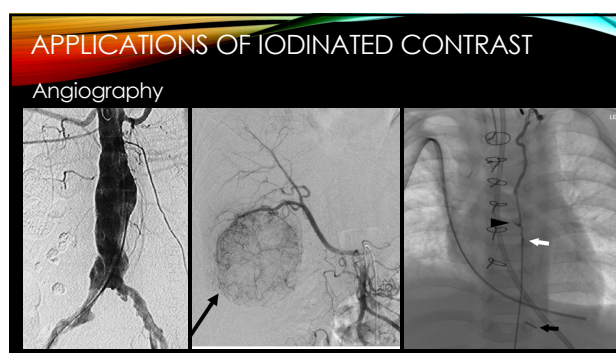
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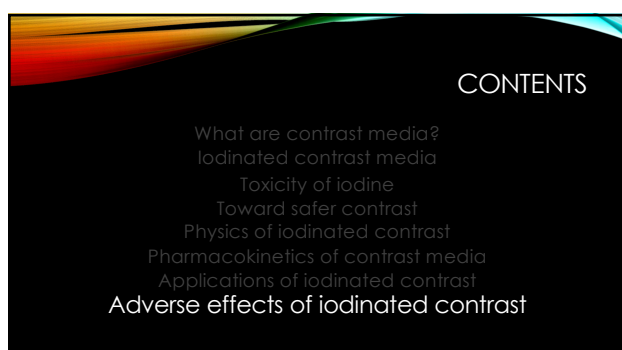
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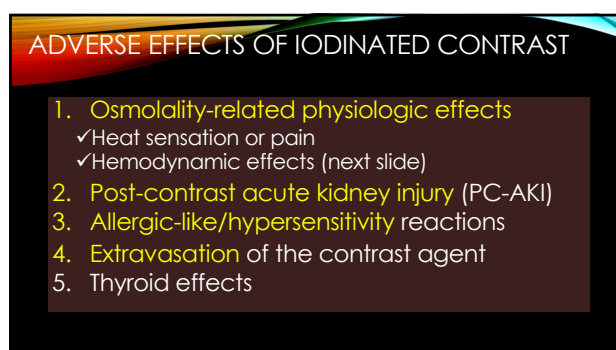
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ADVERSE EFFECTS OF IODINATED CONTRAST

1. Osmolality-related physiologic effects

- Patients requiring particular consideration
 - ✓ Severe cardiovascular disease or heart failure
 - ✓ Dehydration / volume depletion
 - ✓ Neonates and young infants
 - ✓ Shock, particularly with hypovolemia
 - ✓ Other conditions with limited tolerance to rapid fluid shifts

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ADVERSE EFFECTS OF IODINATED CONTRAST

2. Post-contrast acute kidney injury (PC-AKI)

Definition by

*ESUR Contrast Medium Safety Committee:

An increase in serum creatinine ≥ 0.3 mg/dl (or > 26.5 $\mu\text{mol/l}$), or ≥ 1.5 times baseline, within 48-72 hours of intravascular administration of a contrast agent.

PC-AKI = AKI occurring after contrast administration;
association does not establish causality.

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ADVERSE EFFECTS OF IODINATED CONTRAST

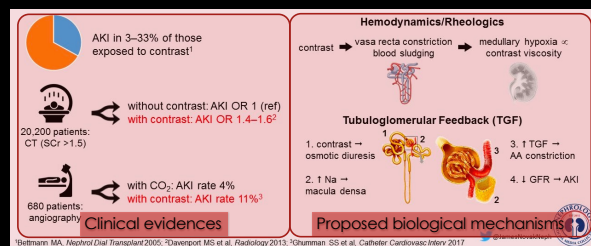
2. Post-contrast acute kidney injury (PC-AKI)

- Patients requiring particular consideration
 - ✓ Acute kidney injury
 - ✓ Severe CKD, especially eGFR < 30 mL/min/1.73 m²
 - ✓ Volume depletion
 - ✓ Severe cardiovascular disease/heart failure when hydration is contemplated
 - ✓ Particular caution with intra-arterial first-pass renal exposure

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ADVERSE EFFECTS OF IODINATED CONTRAST

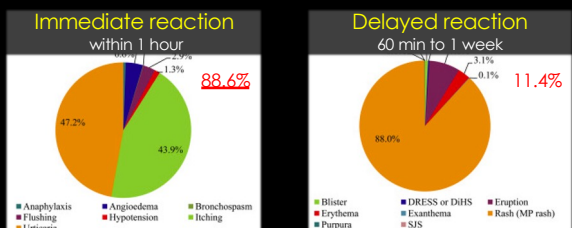
2. Post-contrast acute kidney injury (PC-AKI)



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ADVERSE EFFECTS OF IODINATED CONTRAST

3. Allergic-like/hypersensitivity reactions



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ADVERSE EFFECTS OF IODINATED CONTRAST

3. Allergic-like/hypersensitivity reactions

Risk factors:

- Prior allergic-like reaction to iodinated contrast medium
 - ✓ 5-fold increased risk of developing a future allergic-like reaction to the same class of contrast medium again.

What if your patient has a history of an allergic-like reaction?

1. Identify the culprit agent and severity of the previous reaction
2. Consider an **alternative** examination if appropriate
3. Use a **different** iodinated contrast agent if contrast is necessary
4. **Premedication** may be considered in selected high-risk patients
5. **Be prepared** for breakthrough reactions

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ADVERSE EFFECTS OF IODINATED CONTRAST

3. Allergic-like/hypersensitivity reactions

Risk factors (cont'd):

- **Asthma**
 - ✓ Asthma is associated with an increased risk of allergic-like reactions, particularly bronchospasm.
 - **Allergies unrelated to contrast media**
 - ✓ Two- to three-fold increased risk of an allergic-like contrast reaction.
- ❖ Restricting contrast medium use or premedicating solely on the basis of asthma/unrelated allergies is **NOT recommended**.

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ADVERSE EFFECTS OF IODINATED CONTRAST

3. Allergic-like/hypersensitivity reactions

Risk factors (cont'd):

- **Prior reaction to gadolinium-based (for MRI) contrast media**
 - ✓ **No molecular cross-reactivity** between iodinated and gadolinium-based contrast media.
 - ✓ However, patients with a history of hypersensitivity reactions to gadolinium-based contrast media **may have a general predisposition to reactions**.
- ❖ Again, routine premedication or avoidance is **NOT recommended** solely because of a reaction to contrast from another class

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ADVERSE EFFECTS OF IODINATED CONTRAST

4. Extravasation of the contrast media



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ADVERSE EFFECTS OF IODINATED CONTRAST

4. Extravasation of the contrast media

- The frequency ranges from 0.1% to 1.2% (related to power injection for CT).
- Injection rate is **not significantly associated** with the frequency of extravasation.
- More common when injections are made into more peripherally placed catheters.

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ADVERSE EFFECTS OF IODINATED CONTRAST

4. Extravasation of the contrast media

- The acute tissue injury may occur, in part, related to its hyperosmolality.
- Most patients **recover without clinically important sequelae**.
- Severe injuries, including **compartment syndrome, skin ulceration, or tissue necrosis** are **very uncommon**.

Compartment syndrome

- ✓ Observed after extravasation in **less spacious areas** (such as over the dorsal surface of the wrist).

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SUMMARY

- The contrast agent is **indispensable** in many clinical situations.
- Although there is **no perfect** contrast material, physicians should consider using an appropriate one when clinically indicated.
- **In-depth knowledge** of contrast agents is required to achieve the optimal diagnostic performance while minimizing risk.

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